

Summary of Prim's Algorithm Code

This Python program implements Prim's Algorithm to find the Minimum Spanning Tree (MST) of a weighted, connected graph.

- It starts from vertex 0 and repeatedly selects the vertex with the minimum edge weight.
- `key[]` stores the minimum weight required to connect each vertex to the MST.
- `parent[]` stores the previous vertex used to connect each vertex.
- `mstSet[]` keeps track of vertices already included in the MST.
- Finally, it prints all MST edges and calculates the total minimum cost.

Output: The MST contains edges with a total minimum cost of 16.

Time Complexity: $O(V^2)$

Space Complexity: $O(V^2)$

Output :-



```
Problems  Output  Debug Console  Terminal  Ports
PS C:\Users\shiva\Desktop\UNI Coding\DAA> cd "c:\Users\shiva\Desktop\UNI
Edges of Minimum Spanning Tree:
Edge    weight
0 - 1    2
1 - 2    3
0 - 3    6
1 - 4    5
Total Minimum Cost = 16
PS C:\Users\shiva\Desktop\UNI Coding\DAA>
```